To survive this challenge you need to accumulate at least 20 points.

What is the difference between the cornea and the lens?	What does the cornea look like?	Why is the pupil of the eye more open in low light conditions than in bright light?	What shape is the lens?
3 points	**1 point**	**3 points**	**1 point**
If the muscles in the iris were damaged, what effect would that have on vision?	What is the name of the tough, white outer layer of the eye?	What are the names of the photosensitive cells found in the retina?	What two parts of the eye are responsible for the change in shape of the lens?
4 points	**1 point**	**1 point**	**1 point**
Evaluate the statement, “the eye is like a camera”.	What is the function of the optic nerve?	What is the name given to the part of the eye where the optic nerve meets the retina?	Describe the pupil.
4 points	**2 points**	**1 point**	**2 points**
Why is the eye considered to be a sensory organ?	Explain what a blind spot is.	Write 3 questions the answer to which must be “the retina”.	Sketch and label a diagram of the eye.
2 points	**3 points**	**3 points**	**4 points**
Which parts of the eye need to be transparent?	Pick five of the most important parts of the eye, put them order of most important to least important. Explain why you chosen these parts.	Describe the position of the cornea in the eye.	How do glasses and contact lenses help people to see?
1 point	**4 points**	**2 points**	**3 points**
How many ways could the lens be used? (Don’t just think about inside of the eye).	List ten parts of the eye and put them in alphabetical order.	If you were to build a model of an eye what materials would you use for each part and why?	How could the design of the eye be improved?

3 points		**1 point**		**3 points**		**3 points**	

The cornea is thinner and does not change shape.	Very smooth, transparent. Curved, front part of the sclera.	The muscles in the iris contract and relax depending on the ambient lighting. More light needs to enter the eye when it is dark and so some of the muscles in the iris contract making the pupil dilate.	Bi-convex
The pupils will not change size and therefore in bright conditions, you would not be able to see much due to glare and in low light conditions you may not see anything at all if the pupil is too small.	Sclera	Rods and cones	Ciliary muscle and suspensory ligaments
The eye and camera are similar because both have lens, iris etc. Different because the images from the eye are not captured. Camera is bigger etc.	Carries electrical impulses to the brain	Blind spot	Dark hole in the centre of the iris that can change in size.
It senses light.	Blind spot is where the optic nerve and retina meet. It has no cones or rods so is not sensitive to light.		
Conjunctiva, cornea, lens, vitreous humor		Front of the eye in front of the lens and behind the conjunctiva membrane.	They refract the light so that it is focussed on the retina.

Caused by the light not refracting properly and focusing before or after the retina.			Automatic UV filter built in to protect the eye etc.

Suggested answers